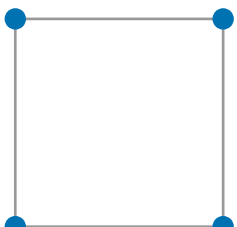
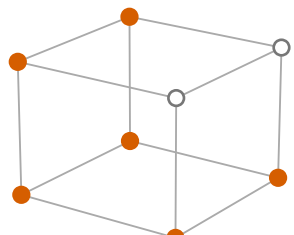
minimal reduced boxes as subsets of AG($k-1, 2$) (filled points)

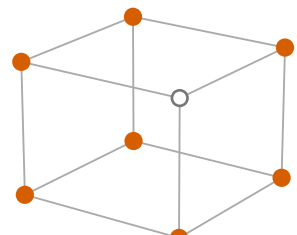
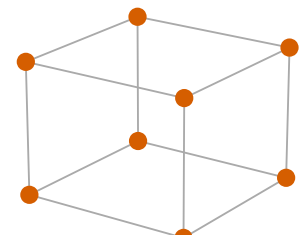
$k = 3$: all of AG(2, 2) (E_4)



$k = 4$, $p = 6$ (28 labelled)



$k = 4$, $p = 7$ (8 labelled)



$k = 4$, $p = 8$ (1 labelled)